

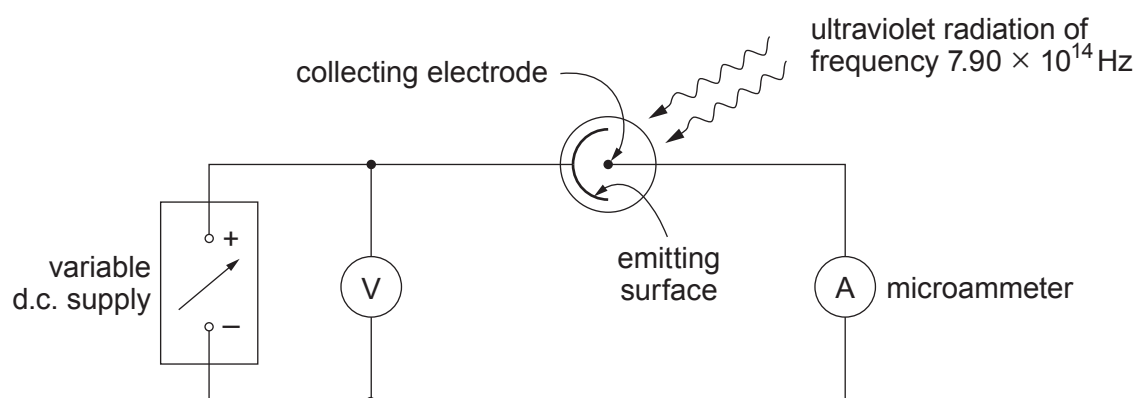
8. (a) (i) State, **in terms of energy**, the meanings of the two right-hand terms in Einstein's photoelectric equation,

$$E_{k\max} = hf - \phi$$

I. hf [1]

II. ϕ [1]

- (ii) Ultraviolet radiation of frequency 7.90×10^{14} Hz is shone on to surfaces of barium and magnesium in turn, using the apparatus shown.



Determine whether or not electrons are emitted from each surface and, if so, the minimum pd needed to reduce the current shown by the microammeter to zero. [3]

[Work function of barium = 4.03×10^{-19} J;
Work function of magnesium = 5.86×10^{-19} J]

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- (b) Photovoltaic panels transfer some of the energy from the Sun into electrical energy. It has been estimated that a quarter of the UK's electrical energy needs could be provided by photovoltaic panels covering an area equal to the total area of roofs on UK buildings. Discuss whether it should be compulsory to have photovoltaic panels on all roofs. [3]

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